

Algebra 1

Unit 1

Lesson 6 Homework

Name **Answer Key** _____

Date _____

Period _____

1. Explain mathematically why the expressions $\left(a^{\frac{1}{c}}\right)^b$ is equivalent to $(a^b)^{\frac{1}{c}}$ using the expressions $\left(-8^{\frac{1}{3}}\right)^2$ and $(-8^2)^{\frac{1}{3}}$.

$$\left(-8^{\frac{1}{3}}\right)^2 = -8^{\frac{1}{3} \cdot 2} = -8^{\frac{2}{3}} = -8^{2 \cdot \frac{1}{3}} = (-8^2)^{\frac{1}{3}}$$

2. Identify all of the expressions equivalent to $\left(27^{\frac{2}{3}}\right)^2$.

- $27^{\frac{3}{4}}$
- $\sqrt[4]{27^3}$
- $(27^2)^{\frac{2}{3}}$ ✓
- $\sqrt[3]{27^4}$ ✓
- $(\sqrt[3]{27})^4$ ✓

For questions 3-8, complete each row with equivalent expressions.

	$a^{\frac{b}{c}}$	$\left(a^{\frac{1}{c}}\right)^b$	$(a^b)^{\frac{1}{c}}$	$(\sqrt[c]{a})^b$	$\sqrt[c]{a^b}$
3.	$25^{\frac{5}{2}}$	$\left(25^{\frac{1}{2}}\right)^5$	$(25^5)^{\frac{1}{2}}$	$(\sqrt{25})^5$	$\sqrt{25^5}$
4.	$(-27)^{\frac{2}{3}}$	$\left[(-27)^{\frac{1}{3}}\right]^2$	$[(-27)^2]^{\frac{1}{3}}$	$(\sqrt[3]{-27})^2$	$\sqrt[3]{(-27)^2}$
5.	$70^{\frac{4}{5}}$	$\left(70^{\frac{1}{5}}\right)^4$	$(70^4)^{\frac{1}{5}}$	$(\sqrt[5]{70})^4$	$\sqrt[5]{70^4}$
6.	$8^{\frac{7}{3}}$	$\left(8^{\frac{1}{3}}\right)^7$	$(8^7)^{\frac{1}{3}}$	$(\sqrt[3]{8})^7$	$\sqrt[3]{8^7}$
7.	$121^{\frac{3}{2}}$	$\left(121^{\frac{1}{2}}\right)^3$	$(121^3)^{\frac{1}{2}}$	$(\sqrt{121})^3$	$\sqrt{121^3}$

For questions 8-10, rewrite each expression using a rational exponent in the form $a^{\frac{b}{c}}$.

8. $\sqrt[3]{(-48)^8}$

$$(-48)^{\frac{8}{3}}$$

9. $(\sqrt{18})^3$

$$18^{\frac{3}{2}}$$

10. $\frac{1}{\sqrt[4]{10^5}}$

$$\frac{1}{10^{\frac{5}{4}}} = 10^{-\frac{5}{4}}$$